SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA17**

COURSE OUTCOME COVERED

CO3: *Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)*

Assessment Tool Weightage: 50%

Time Duration: 1 Hour

QUESTIONS: 4 **Total Marks:40**

Annexure A

CASE STUDY

Code of Conduct:

I ____________________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

CASE STUDY

Case Study 1 — Smart Medical Diagnosis System

A hospital is designing a rule-based Logical Agent for preliminary patient diagnosis. The system's knowledge base contains:

I. If a patient has Fever AND Cough → Possible Flu
Ii. If a patient has Fever AND Rash → Possible Measles
Iii. If a patient has Cough AND Breathlessness → Possible Pneumonia
Iv. Facts about Patient A: Fever = True, Cough = True, Breathlessness = True

(a) Represent the above rules and facts precisely using Propositional Logic symbols. Define each symbol used.

(b) Apply Forward Chaining on the knowledge base to derive all possible diagnoses for Patient A. Show each inference step with the rule fired and fact derived.

(c) The goal is to confirm 'Patient A has Pneumonia.' Apply Backward Chaining and draw the goal reduction tree to verify this goal.

Case Study 2 — Autonomous Traffic Management Agent

A smart city traffic controller uses a FOL-based knowledge base to manage emergency vehicle routing:

I. ∀x: Vehicle(x) ∧ EmergencyType(x) → ClearPath(x) [Rule 1]
II. ∀x: ClearPath(x) → GreenSignal(x) ∧ ¬RedSignal(x) [Rule 2]
III. ∀x ∀y: Vehicle(x) ∧ Behind(x,y) ∧ GreenSignal(y) → Proceed(x) [Rule 3]
IV. Facts: Vehicle(Ambulance), EmergencyType(Ambulance), Vehicle(CarA), Behind(CarA, Ambulance)

(a) Apply Unification to match Rule 1 with the given facts. Show the complete substitution set θ (MGU) at each step.

(b) Using the Resolution method, prove that 'Proceed (CarA)' is true. Convert relevant clauses to CNF and show each resolution step.

(c) Analyze whether the current FOL knowledge base is sufficient to handle two simultaneous emergency vehicles. Identify what additional rules or facts are needed and justify logically.

Case Study 3 — Agricultural Expert Reasoning System

An agricultural advisory logical agent operates with the following propositional knowledge base (KB):

1) P1: SoilDry → IrrigationNeeded
2) P2: IrrigationNeeded ∧ CropWheat → ApplyDripMethod
3) P3: ¬ApplyDripMethod ∧ CropWheat → CropAtRisk
4) Facts: SoilDry = True, CropWheat = True

(a) Convert all sentences P1, P2, P3 and the facts into Conjunctive Normal Form (CNF). Show each conversion step (eliminate implication → move ¬ inward → distribute).

(b) Using Resolution Refutation, prove the theorem 'ApplyDripMethod'. Negate the goal, combine with the KB in CNF, and derive the empty clause □. Draw the resolution proof tree.

(c) If the fact 'SoilDry' is removed from the KB, analyze step-by-step how logical conclusions change. Does 'CropAtRisk' still follow? Apply appropriate reasoning patterns to justify.

Case Study 4 — Wumpus World Knowledge Agent

An AI agent navigating the Wumpus World perceives: Stench at [1,2], Breeze at [1,1], Glitter at [2,2]. The agent uses a propositional KB with these rules:

1) Stench [1,2] → Wumpus is adjacent to cell [1,2]
2) Breeze [1,1] → Pit is adjacent to cell [1,1]
3) Glitter [x,y] → Gold is at [x,y]
4) ¬Wumpus [x,y] ∧ ¬Pit[x,y] → Safe[x,y]

(a) Construct the agent's belief state from its percepts. Apply Modus Ponens systematically to derive new knowledge facts. List all derived conclusions.

(b) Using Forward Chaining from the agent's percepts, identify the possible locations of the Wumpus and Pit. Show each reasoning step precisely.

(c) Evaluate whether the path [1,1] → [1,2] → [2,2] is safe for the agent to collect the gold. Use the knowledge derived in (a) and (b) to justify your conclusion with logical inference.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand

CASE STUDY 1 — Smart Medical Diagnosis System

(a) Representation using Propositional Logic

Let the following propositions represent the patient's conditions:

- **F** = Patient has Fever
- **C** = Patient has Cough
- **B** = Patient has Breathlessness
- **Flu** = Patient has Possible Flu
- **M** = Patient has Possible Measles
- **P** = Patient has Possible Pneumonia

Rules

Rule I:

If a patient has Fever AND Cough → Possible Flu

Rule II:

If a patient has Fever AND Rash → Possible Measles
Let **R** = Patient has Rash.

Rule III:

If a patient has Cough AND Breathlessness → Possible Pneumonia

Facts about Patient A
Therefore, the initial knowledge base is:
where:

- = AND
- = implication
- True = fact is known to be true
- A = Patient A
-

(b) Forward Chaining

Forward chaining starts with the **known facts** and repeatedly applies rules whose conditions are satisfied.

Initial Facts

Step 1 — Apply Rule I

Rule:
We know:
Therefore:

Derived fact: Patient A has Possible Flu.

Step 2 — Apply Rule II

Rule:
We know:
But there is **no fact that Patient A has Rash**.
Therefore:
Hence Rule II **cannot fire**.
So:

Step 3 — Apply Rule III

Rule:
We know:
Therefore:

Derived fact: Patient A has Possible Pneumonia.
Forward-Chaining Result

Step	Rule	Conditions	New Fact
0	Facts	F, C, B	Initial facts
1	Rule I	F ∧ C	Flu
2	Rule II	F ∧ R	Cannot fire; R unknown
3	Rule III	C ∧ B	Pneumonia

Final conclusions
Measles cannot be concluded because **Rash is not given**.

(c) Backward Chaining for Pneumonia
Goal
Backward chaining starts from the goal and searches for a rule that can produce it.
The only rule that concludes Pneumonia is:
Therefore, the goal is reduced to two sub-goals:
↓ Rule III
Now check the facts:
Both sub-goals are satisfied.

CASE STUDY 2 — Autonomous Traffic Management Agent

(a) Unification with Rule 1

Rule 1
Given facts:
We need to match the rule variable with the constant **Ambulance**.

Step 1
Match:
with:
Therefore:
Step 2
Check the second condition:
Substituting :
This is already a fact.

MGU
Therefore, the Most General Unifier is:
After substitution:
Hence:
is derived.

(b) Resolution Proof of Proceed(CarA)
We want to prove:
Resolution uses **proof by contradiction**, so add the negated goal:
Step 1 — Convert Rule 1 to CNF
Original:
Eliminate implication:
Therefore:

Step 2 — Rule 2
Original:
This produces two clauses:

Step 3 — Rule 3
Original:
CNF:

Step 4 — Facts
Add negated goal:

Resolution Step 1
Using with and :
Since both positive facts are present:

Resolution Step 2
Use :
With:
we obtain:

Resolution Step 3
Instantiate Rule 3 with:
Thus:
We have:
Therefore:

Resolution Step 4 — Contradiction
We also have the negated goal:
Resolve:
with:
Result:
where is the **empty clause**, indicating contradiction.
Therefore:
Hence, **CarA is logically proven to proceed**.

(c) Handling Two Simultaneous Emergency Vehicles
The current KB can represent more than one vehicle because the rules use universally quantified variables:
Therefore, Rule 1 can independently derive:
and
if both satisfy the emergency conditions.
However, the KB is **not completely sufficient for realistic simultaneous emergency management**.

Missing information
It should include facts such as:
It also needs spatial relationships:

Additional useful rule
A priority rule can be added:
Another rule can handle conflicting emergency vehicles

Analysis
The existing KB establishes **individual emergency clearance**, but does not specify:

- which emergency vehicle gets priority when routes conflict,
- how two emergency vehicles are separated,
- what happens when both require the same intersection,
- how ordinary vehicles should react to multiple emergency vehicles.

Therefore, additional **priority, conflict-resolution, and routing rules** are required.

CASE STUDY 3 — Agricultural Expert Reasoning System

(a) CNF Conversion

P1
Original:
Eliminate implication:
Therefore:
This is already CNF.

P2
Original:
Eliminate implication:
Apply De Morgan's law:
Therefore:

P3
Original:
Eliminate implication:
Using De Morgan's law:
Therefore:

Facts
CNF:
and
CNF:
Complete CNF Knowledge Base

(b) Resolution Refutation for ApplyDripMethod
Goal
Negate the goal:
Add this to the KB.

Resolution 1
We have:
and:
Resolve them:

Resolution 2
Use:
We have:
and:
Therefore:

Resolution 3

We also added:
Resolve:
with:
Therefore:
The empty clause proves the contradiction.
Hence:

Resolution Proof Tree

```
        SoilDry
          |
          | P1
          ↓
     IrrigationNeeded
          |
          | P2 + CropWheat
          ↓
     ApplyDripMethod
          |
          | Negated Goal
          ↓
    ¬ApplyDripMethod
          |
          ↓
          □
```

Thus, **ApplyDripMethod is logically entailed by the KB**.

(c) Removing SoilDry

Suppose:
is removed.
The remaining facts are:
There is now no fact that establishes:
Therefore, P1:
cannot be fired.

Step 1

Without SoilDry:
cannot be derived.

Step 2

P2 requires:
Although:
we do not have:
Therefore:
cannot be derived.

Step 3 — CropAtRisk

P3 requires:
We have:

but we **do not have**:
Therefore:
also cannot be derived.
Important logical point
The absence of:
does **not** mean:
This is the **open-world reasoning principle**.
Therefore:

Final comparison

Knowledge	With SoilDry	Without SoilDry
SoilDry	True	Unknown
IrrigationNeeded	Derived	Not derived
CropWheat	True	True
ApplyDripMethod	Derived	Not derived
¬ApplyDripMethod	Not given	Not given
CropAtRisk	Not derived	Not derived

Conclusion: Removing SoilDry breaks the inference chain leading to IrrigationNeeded and ApplyDripMethod. However, CropAtRisk still does **not** follow because its required negative premise, , is unavailable.

CASE STUDY 4 — Wumpus World Knowledge Agent

(a) Belief State and Modus Ponens
Given percepts
The agent perceives:

1. Stench at [1,2]
2. Breeze at [1,1]
3. Glitter at [2,2]

Therefore the initial belief state is:

Step 1 — Stench
Rule:
Given:
By Modus Ponens:

Step 2 — Breeze
Rule:
Given:
By Modus Ponens:

Step 3 — Glitter

Rule:
For:
we obtain:
Therefore:

Derived Belief State
The agent can therefore conclude that **gold is definitely at [2,2]**, while the exact Wumpus and pit locations require further reasoning.

(b) Forward Chaining for Wumpus and Pit Locations
Assume the standard Wumpus World adjacency relationship: cells sharing a side are adjacent.
Wumpus
We know:
The adjacent cells to [1,2] are:
Therefore:
This means the current KB does **not identify one unique Wumpus location**.

Pit
We know:
The adjacent cells to [1,1] are:
Therefore:
Again, the exact location cannot be determined from the available information.
Reasoning Summary

```
Stench[1,2]
     |
     ↓
Wumpus is adjacent to [1,2]
     |
     ├── [1,1]
     ├── [1,3]
     └── [2,2]


Breeze[1,1]
     |
     ↓
Pit is adjacent to [1,1]
     |
     ├── [1,2]
     └── [2,1]
```

Thus:

(c) Safety of Path [1,1] → [1,2] → [2,2]
The proposed path is:

We need to check every cell.
Cell [1,1]
No direct fact establishes:
or:
However, the simplified rule only establishes that a pit is **adjacent** to [1,1]. It does not establish that [1,1] itself contains a pit.
So [1,1] is **not proven dangerous**.

Cell [1,2]
This cell is problematic.
From:
we know that a pit is adjacent to [1,1]:
Therefore [1,2] is a **possible pit location**.
Also, because:
we know a Wumpus is adjacent to [1,2], although the simplified rule does not establish that the Wumpus is actually at [1,2].
Thus [1,2] cannot be proven safe.

Cell [2,2]
We know:
which means the desired gold is there.
However, the Wumpus rule allows [2,2] to be one of the possible Wumpus locations because [2,2] is adjacent to [1,2]:
Therefore [2,2] is also **not proven safe** from the available knowledge.

Safety Rule
The KB contains:
To derive:
we must know both:
and:
But the KB provides neither of these negative facts for the cells on the proposed path.
Therefore, we cannot derive:
or:

Final Decision
In particular, **[1,2] should not be treated as safe**, because it is a possible pit location.
The agent should therefore **not blindly follow the path** with the current knowledge. It should first obtain additional percepts or knowledge that eliminates the dangerous possibilities.